

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular/Supplementary Winter Examination – 2024

Course: B. Tech.

Semester :VII

Branch: Electronics & Telecommunication Engineering /Electronics and Communication Engineering

Subject Code & Name: BTETC701 Microwave Engineering

Max Marks: 60

Date:05/02/2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	The modes of propagation supported by a rectangular wave guide is: a. TM, TEM, TE b. TM, TE c. TM, TEM d. TE, TEM	CO1	1
2	The Smith chart is based on the polar plot of a. Voltage reflection coefficient b. Impedance c. Reactance d. Current	CO1	1
3	A cylindrical cavity resonator can be constructed using a circular waveguide. a. shorted at both the ends b. open at both the ends c. matched at both the ends d. none of the mentioned	CO1	1
4	The device used to get the measurement of S parameters of n- port micro wave network is a. CRO b. Network analyzer c. Attenuator d. Circulator	CO2	1
5	Ferrite isolators are ____ port microwave devices. a. Eight b. Three c. Two d. Four	CO2	1
6	Which device is based on Faraday rotation? a. Magic Tee b. E plane Tee c. Isolator d. H plane tee	CO2	1
7	_____ is a single cavity klystron tube that operates as on oscillator by using a reflector electrode after the cavity. a. Backward wave oscillator b. Reflex klystron c. Travelling wave tube d. Magnetrons	CO3	1
8	Klystron operates on the principle of a. Amplitude Modulation b. Frequency Modulation c. Pulse Modulation d. Velocity Modulation	CO3	1
9	The range of VSWR in dB indicated in the VSWR meter is ____ a. 0–2 dB b. 0–10 dB c. 0–5 dB d. 0–100 dB	CO4	1
10	The Double Minimum method is relevant to the measurement of	CO4	1

	a. low attenuation coefficient	b. high attenuation coefficient	c. high VSWR	d. low VSWR		
11	Stripline can be compared to a:				CO5	1
	a. Flattened rectangular waveguide	b. Flattened circular waveguide	c. Flattened coaxial cable	d. None of the mentioned		
12	The mode of propagation in a microstrip line is				CO5	1
	a. Quasi TEM	b. TEM	c. TM	d. TE		
Q.2	Solve the following.					12
A)	An air filled rectangular waveguide of inside dimension 7 X 3.5 cm, operates in dominant TE ₁₀ mode. Find 1. Cut off Frequency 2. Determine the phase velocity of wave in guide at 3.5 GHz. 3. Determine the guided wavelength at same frequency.				CO1	6
B)	Compare transmission lines and waveguides.				CO1	6
Q.3	Solve the following.					12
A)	Explain E- plane Tee with help of S matrix.				CO2	6
B)	Explain working principal of directional coupler with help of neat and labeled diagram. State and explain its performance parameters.				CO2	6
Q.4	Solve Any Two of the following.					12
A)	Explain Construction and Principle of operation of helix type travelling wave tube (TWT) in detail.				CO3	6
B)	Explain Reflex klystron with help of Applegate diagram.				CO3	6
C)	Explain Construction and Principle of operation of 8 cavity cylindrical travelling wave magnetron.				CO3	6
Q.5	Solve Any Two of the following.					12
A)	Explain Impedance measurement techniques at microwave frequency.				CO4	6
B)	Define VSWR. Explain the VSWR meter with the help of diagram.				CO4	6
C)	Explain the power measurement technique with the help of diagram.				CO4	6
Q.6	Solve Any Two of the following.					12
A)	Explain the micro strip lines in detail with neat diagram. State its applications.				CO5	6
B)	Explain Structural details and applications of Stripline.				CO5	6
C)	What are the hazards of Electromagnetic Radiation?				CO5	6
*** End ***						